

11.1 Basic Concepts

1. Describe the difference between $(4, -1)$ and $\langle 4, -1 \rangle$. Illustrate the difference with a sketch.

For problems 2 - 6 give the vector for the set of points. Find its magnitude and determine if the vector is a unit vector.

2. The line segment from $(6, -2, 3)$ to $(-3, -2, 1)$.
3. The line segment from $\left(1, -\frac{1}{5}\right)$ to $\left(\frac{8}{5}, \frac{3}{5}\right)$.
4. The line segment from $(2, -1, 5)$ to $(8, -3, -6)$.
5. The position vector for $(-12, 4, 8)$.
6. The position vector for $(\cos(\theta), \sin(\theta))$ for any angle θ .
7. The vector $\vec{v} = \langle -8, -3 \rangle$ starts at the point $P = (8, 2)$. At what point does the vector end?
8. The vector $\vec{v} = \langle 0, 5, -3 \rangle$ starts at the point $P = (-1, 0, 5)$. At what point does the vector end?
9. The vector $\vec{v} = \langle -8, -3 \rangle$ ends at the point $P = (8, 2)$. At what point does the vector start?
10. The vector $\vec{v} = \langle 4, -2, 1 \rangle$ ends at the point $P = (7, -7, 2)$. At what point does the vector start?

11.2 Vector Arithmetic

- Given $\vec{a} = 3\vec{i} - 9\vec{j}$ and $\vec{b} = -6\vec{i} + \vec{j}$ compute each of the following.
 - $10\vec{b}$
 - $14\vec{a} + 20\vec{b}$
 - $\left\| 8\vec{b} - \frac{1}{3}\vec{a} \right\|$
- Given $\vec{u} = \langle 0, 4, -1 \rangle$ and $\vec{v} = \langle 6, -2, -7 \rangle$ compute each of the following.
 - $\frac{3}{4}\vec{u}$
 - $-3\vec{u} - 7\vec{v}$
 - $\|\vec{v} + 10\vec{u}\|$
- Given $\vec{p} = \langle 3, -1, -2 \rangle$ and $\vec{q} = -\frac{1}{3}\vec{i} - \frac{1}{2}\vec{k}$ compute each of the following.
 - $2\vec{p}$
 - $9\vec{q} - 2\vec{p}$
 - $\|8\vec{p} - 12\vec{q}\|$
- Find a unit vector that points in the same direction as $\vec{a} = \langle 10, -3, 8, -2 \rangle$.
- Find a unit vector that points in the same direction as $\vec{w} = -\vec{i} - 6\vec{j}$.
- Find a unit vector that points in the opposite direction as $\vec{c} = 2\vec{i} + 7\vec{j} - 5\vec{k}$.
- Find a unit vector that points in the opposite direction as $\vec{b} = \langle 0, -3, -11 \rangle$.
- Find a vector that points in the same direction as $\vec{p} = 2\vec{i} - 3\vec{j} + \vec{k}$ with a magnitude of $\frac{1}{2}$.
- Find a vector that points in the opposite direction as $\vec{a} = \langle -3, -14, 2 \rangle$ with a magnitude of 32.
- Find a vector that points in the same direction as $\vec{b} = -3\vec{i} + 2\vec{k}$ with a magnitude that is $\frac{1}{10}$ the magnitude of $\vec{b}$.
- Determine if $\vec{p} = 8\vec{i} - 3\vec{j}$ and $\vec{q} = 16\vec{i} - 6\vec{j}$ are parallel vectors.
- Determine if $\vec{v} = \langle 1, 0, -4 \rangle$ and $\vec{w} = \langle 9, 3, 1 \rangle$ are parallel vectors.
- Determine if $\vec{a} = 10\vec{i} + 8\vec{j} + 20\vec{k}$ and $\vec{b} = \langle -35, -28, 70 \rangle$ are parallel vectors.

14. Prove the property : $\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}$.
15. Prove the property : $\vec{v} + \vec{0} = \vec{v}$.
16. Prove the property : $1\vec{v} = \vec{v}$.
17. Prove the property : $(a + b)\vec{v} = a\vec{v} + b\vec{v}$.



11.3 Dot Product

For problems 1 - 5 determine the dot product, $\vec{a} \cdot \vec{b}$.

1. $\vec{a} = 9\vec{i} - 8\vec{k}$, $\vec{b} = \langle 3, -2, 1 \rangle$

2. $\vec{a} = \langle 4, -1, 0, 5 \rangle$, $\vec{b} = \langle 3, 0, -10, 6 \rangle$

3. $\vec{a} = \vec{i} - 5\vec{j} - 2\vec{k}$, $\vec{b} = -4\vec{i} + 2\vec{j} + 8\vec{k}$

4. $\|\vec{a}\| = \frac{1}{4}$, $\|\vec{b}\| = \frac{9}{4}$ and the angle between the two vectors is $\theta = \pi$.

5. $\|\vec{a}\| = 24$, $\|\vec{b}\| = 9$ and the angle between the two vectors is $\theta = \frac{2\pi}{7}$.

For problems 6 - 8 determine the angle between the two vectors.

6. $\vec{p} = 9\vec{i} - \vec{j}$, $\vec{q} = -3\vec{i} - 6\vec{j}$

7. $\vec{a} = \langle 4, 0, -3 \rangle$, $\vec{b} = 2\vec{i} + 10\vec{j} - 11\vec{k}$

8. $\vec{w} = \langle 8, 3, -1, -4 \rangle$, $\vec{v} = \langle -1, 9, 4, -8 \rangle$

For problems 9 - 12 determine if the two vectors are parallel, orthogonal or neither.

9. $\vec{q} = 7\vec{i} - 14\vec{j} - 21\vec{k}$, $\vec{p} = \langle -4, 8, 12 \rangle$

10. $\vec{u} = \langle 5, 0, -2 \rangle$, $\vec{q} = \langle 4, -7, 10 \rangle$

11. $\vec{a} = 9\vec{i} - \vec{j} + 5\vec{k}$, $\vec{b} = -2\vec{i} + 7\vec{j} + \vec{k}$

12. $\vec{v} = \langle -1, 3, 1, 5 \rangle$, $\vec{w} = \langle -8, 3, -7, -2 \rangle$

13. Given that $\vec{a} \cdot \vec{b} = -6$, $\|\vec{a}\| = 4.3$ and the angle between $\vec{a}$ and $\vec{b}$ is $\theta = \frac{\pi}{6}$ determine if $\vec{b}$ is a unit vector or not.

For problems 14 & 15 determine the value of b for which the two vectors will be orthogonal.

14. $\vec{u} = \langle 3, -1, 6 \rangle$, $\vec{v} = \langle 3, -2b, 1 \rangle$

15. $\vec{u} = \langle 1 - b, 4, -2 \rangle$, $\vec{v} = \langle b, 6, 3b \rangle$

16. Given $\vec{a} = \vec{i} + 3\vec{j} - 2\vec{k}$ and $\vec{b} = -3\vec{i} - 4\vec{j} + 7\vec{k}$ compute $\text{proj}_{\vec{a}} \vec{b}$.

17. Given $\vec{a} = \vec{i} + 3\vec{j} - 2\vec{k}$ and $\vec{b} = -3\vec{i} - 4\vec{j} + 7\vec{k}$ compute $\text{proj}_{\vec{b}} \vec{a}$.

18. Given $\vec{p} = \langle 5, -2, 1 \rangle$ and $\vec{q} = \langle 0, 4, 8 \rangle$ compute $\text{proj}_{\vec{p}} \vec{q}$.
19. Given $\vec{u} = \langle 1, 3, 0, -2 \rangle$ and $\vec{w} = \langle -2, 2, 4, 1 \rangle$ compute $\text{proj}_{\vec{w}} \vec{u}$.
20. Determine the direction cosines and direction angles for $\vec{r} = \langle 5, 2, -7 \rangle$.
21. Determine the direction cosines and direction angles for $\vec{r} = \left\langle \frac{1}{2}, -\frac{3}{4}, \frac{5}{2} \right\rangle$.
22. Prove the property $(c\vec{v}) \cdot \vec{w} = \vec{v} \cdot (c\vec{w})$.
23. Prove the property $\vec{v} \cdot \vec{w} = \vec{w} \cdot \vec{v}$.
24. Prove the property $\vec{v} \cdot \vec{0} = 0$.
25. Prove the property $\vec{v} \cdot \vec{v} = \|\vec{v}\|^2$.



11.4 Cross Product

1. If $\vec{w} = \langle 1, 0, -3 \rangle$ and $\vec{v} = \langle 6, -3, -4 \rangle$ compute $\vec{v} \times \vec{w}$.
2. If $\vec{w} = \langle 1, 0, -3 \rangle$ and $\vec{v} = \langle 6, -3, -4 \rangle$ compute $\vec{w} \times \vec{v}$.
3. If $\vec{a} = 3\vec{i} - 2\vec{j} + 6\vec{k}$ and $\vec{b} = \langle 4, -1, -6 \rangle$ compute $\vec{a} \times \vec{b}$.
4. Find a vector that is orthogonal to the plane containing the points $P = (-4, 2, 6)$, $Q = (-3, 2, 1)$ and $R = (2, -1, 1)$.
5. Find a vector that is orthogonal to the plane containing the points $P = (-1, 1, 6)$, $Q = (-2, 3, 2)$ and $R = (-2, 4, 5)$.
6. Are the vectors $\vec{u} = \langle -2, 4, -1 \rangle$, $\vec{v} = \langle 5, -2, -1 \rangle$ and $\vec{w} = \langle 3, 4, -3 \rangle$ are in the same plane?
7. Are the vectors $\vec{u} = \langle 1, -1, 4 \rangle$, $\vec{v} = \langle 4, 2, -2 \rangle$ and $\vec{w} = \langle -5, 4, -17 \rangle$ are in the same plane?
8. Determine the value of b so that the vectors $\vec{u} = \langle 4, -5, 3 \rangle$, $\vec{v} = \langle -2, 0, -5 \rangle$ and $\vec{w} = \langle b, -1, 6 \rangle$ are in the same plane.

12.6 Vector Functions

For problems 1 - 3 find the domain of the given vector function.

1. $\vec{r}(t) = \left\langle \frac{1}{t^2 - 1}, \frac{1}{t + 3}, \frac{1}{t - 6} \right\rangle$

2. $\vec{r}(t) = \langle \sqrt{t}, \sqrt{t + 1}, \sqrt{t + 2} \rangle$

3. $\vec{r}(t) = \langle \ln(t + 7), \ln(t - 3) \rangle$

For problems 4 - 8 sketch the graph of the given vector function.

4. $\vec{r}(t) = \langle -4, t + 1 \rangle$

5. $\vec{r}(t) = \langle -2 \cos(t), 5 \sin(t) \rangle$

6. $\vec{r}(t) = \langle \sqrt{t + 2}, 1 - t \rangle$

7. $\vec{r}(t) = \langle 2t + 1, t^2 - 1 \rangle$

8. $\vec{r}(t) = \langle t^2 + 4, 6 - t^2 \rangle$

For problems 9 - 12 identify the graph of the vector function without sketching the graph.

9. $\vec{r}(t) = \langle 6, 2 + 8t, -1 + 10t \rangle$

10. $\vec{r}(t) = \langle 12t, 6 - 8t, 4 + 7t \rangle$

11. $\vec{r}(t) = \langle 2, 6 \cos(t), 6 \sin(t) \rangle$

12. $\vec{r}(t) = \langle -2t, 6 \cos(t), 6 \sin(t) \rangle$

For problems 13 - 16 write down the equation of the line segment between the two points.

13. The line segment starting at $(4, -7)$ and ending at $(2, 0)$.

14. The line segment starting at $(-1, 2)$ and ending at $(7, -2)$.

15. The line segment starting at $(4, 1, -3)$ and ending at $(-1, 2, 6)$.

16. The line segment starting at $(1, -1, 9)$ and ending at $(4, -7, 10)$.

12.7 Calculus with Vector Functions

For problems 1 - 6 evaluate the given limit.

$$1. \lim_{t \rightarrow 0} \left(\cos(2t) \vec{i} - e^{4-t} \vec{j} + (t^2 + 3t - 9) \vec{k} \right)$$

$$2. \lim_{t \rightarrow 4} \left\langle \frac{t-4}{t^2-3t-4}, \frac{t^2-4t}{16-t^2} \right\rangle$$

$$3. \lim_{t \rightarrow 0} \left\langle \frac{\sin(t)}{2t}, \frac{1-\cos(t)}{t}, -3 \right\rangle$$

$$4. \lim_{t \rightarrow -8} \left(\frac{e^{t^2-64} - 1}{t+8} \vec{i} + \frac{\sin(t+8)}{t+8} \vec{j} - \vec{k} \right)$$

$$5. \lim_{t \rightarrow -\infty} \left\langle \frac{5t^2-8t+1}{12+5t^2}, \frac{2+t^3}{1+t^2+t^4} \right\rangle$$

$$6. \lim_{t \rightarrow \infty} \left\langle \ln\left(1 - \frac{4}{t}\right), e^{\frac{1}{t^2}}, 2 \right\rangle$$

For problems 7 - 11 compute the derivative of the given vector function.

$$7. \vec{r}(t) = \left\langle \sqrt{3t}, \frac{1}{t^4}, \frac{1}{2t} \right\rangle$$

$$8. \vec{r}(t) = \cos(2t) \vec{i} - \sin(2t) \vec{j} + \ln(2t) \vec{k}$$

$$9. \vec{r}(t) = \left\langle e^{t^2-1}, 4 - \sec(2t), 7 \right\rangle$$

$$10. \vec{r}(t) = \sin(t) \cos(t) \vec{i} - t^3 \ln(t^2) \vec{j}$$

$$11. \vec{r}(t) = \left\langle \frac{1}{(t^2-4)^2}, \frac{t^3}{t^2+2}, \frac{t^2+2}{t^3} \right\rangle$$

For problems 12 - 15 evaluate the given integral.

$$12. \int \vec{r}(t) dt, \text{ where } \vec{r}(t) = \left\langle \frac{1}{t^3} - t^3, 5t, \frac{1}{6t} - \frac{8}{t^4} \right\rangle$$

$$13. \int \vec{r}(t) dt, \text{ where } \vec{r}(t) = (t^2 - 5) e^{t^3-15t} \vec{i} + 4t\sqrt{t^2+1} \vec{j} - \sin^2(5t) \vec{k}$$

$$14. \int_0^1 \vec{r}(t) dt \text{ where } \vec{r}(t) = \langle t \cos(\pi t), 8t - 2, 12t^3 - e^{2t} \rangle$$

15. $\int_1^{-1} \vec{r}(t) dt$ where $\vec{r}(t) = \tan(t) \vec{i} - \sin^3(t) \cos^2(t) \vec{j} - 8t$



12.8 Tangent, Normal and Binormal Vectors

For problems 1 - 3 find the unit tangent vector for the given vector function.

1. $\vec{r}(t) = t^2 \vec{i} - \cos(8t) \vec{j} + \sin(8t) \vec{k}$

2. $\vec{r}(t) = \langle 8t, 2 - t^6, t^4 \rangle$

3. $\vec{r}(t) = \langle \ln(6t), e^{1-t}, 5t \rangle$

For problems 4 & 5 find the tangent line to the vector function at the given point.

4. $\vec{r}(t) = \langle 3 + t^2, t^4, 6 \rangle$ at $t = -1$.

5. $\vec{r}(t) = \langle 2t, \cos^2(t), e^{6t} \rangle$ at $t = 0$.

For problems 6 & 7 find the unit normal and the binormal vectors for the given vector function.

6. $\vec{r}(t) = \langle e^{4t} \sin(t), e^{4t} \cos(t), 2 \rangle$

7. $\vec{r}(t) = 2t \vec{i} + \frac{1}{2}t^2 \vec{j} + \ln(t^2) \vec{k}$

12.9 Arc Length with Vector Functions

For problems 1 - 3 determine the length of the vector function on the given interval.

1. $\vec{r}(t) = 4 \cos(2t) \vec{i} + 3t \vec{j} - 4 \sin(2t) \vec{k}$ from $0 \leq t \leq 3\pi$.

2. $\vec{r}(t) = \langle 9 - 2t, 4 + 2t, \sqrt{2}t^2 \rangle$ from $0 \leq t \leq 1$.

3. $\vec{r}(t) = 2t \vec{i} + \frac{1}{2}t^2 \vec{j} + \ln(t^2) \vec{k}$ from $1 \leq t \leq 3$.

For problems 4 - 6 find the arc length function for the given vector function.

4. $\vec{r}(t) = \langle 8t, 6 + t, -7t \rangle$

5. $\vec{r}(t) = \langle 8t, 4t^{\frac{3}{2}}, 3 \rangle$

6. $\vec{r}(t) = \langle e^{4t} \sin(t), e^{4t} \cos(t), 2 \rangle$

7. Determine where on the curve given by $\vec{r}(t) = \langle 8t, 4t^{\frac{3}{2}}, 3 \rangle$ we are after traveling a distance of 4.

8. Determine where on the curve given by $\vec{r}(t) = \langle e^{4t} \sin(t), e^{4t} \cos(t), 2 \rangle$ we are after traveling a distance of 15.

12.10 Curvature

Find the curvature for each the following vector functions.

1. $\vec{r}(t) = \langle 5t, 1 - 2t, 4t^{\frac{3}{2}} \rangle$

2. $\vec{r}(t) = \langle 6, e^{-5t}, 3te^{-5t} \rangle$

3. $\vec{r}(t) = \langle \cos(\omega t), t, \sin(\omega t) \rangle$



12.11 Velocity and Acceleration

1. An objects acceleration is given by $\vec{a} = \cos(2t)\vec{i} + 4t^3\vec{j} + 6\sin(3t)\vec{k}$. The objects initial velocity is $\vec{v}(0) = 6\vec{i} + 2\vec{j} + 7\vec{k}$ and the objects initial position is $\vec{r}(0) = \vec{i} - 9\vec{j} + 6\vec{k}$. Determine the objects velocity and position functions.
2. An objects acceleration is given by $\vec{a} = 10t\vec{i} - 6\vec{j} + t\cos(t)\vec{k}$. The objects initial velocity is $\vec{v}(0) = -\vec{i} + 11\vec{j} - \vec{k}$ and the objects initial position is $\vec{r}(0) = 4\vec{i} + \vec{j} + 10\vec{k}$. Determine the objects velocity and position functions.
3. Determine the tangential and normal components of acceleration for the object whose position is given by $\vec{r}(t) = \langle 5t, 1 - 2t, 4t^{\frac{3}{2}} \rangle$.
4. Determine the tangential and normal components of acceleration for the object whose position is given by $\vec{r}(t) = \langle 6, e^{-5t}, 3te^{-5t} \rangle$.